

Annual Progress Report - Year 2

NSF Award BIO-2024-07821

Project Title: Alpine Ecosystem Response to Climate Change in the Northern Rockies

Principal Investigator: Dr. Sarah Chen

Institution: University of Idaho

Reporting Period: September 1, 2026 - August 31, 2027

Award Amount: \$485,000

Report Submission Date: October 15, 2027

1. Accomplishments

Major Activities:

- Completed second full year of seasonal monitoring across all 24 permanent plots.
- Deployed 6 additional soil moisture sensors at elevations above 2,800m.
- Conducted drone-based multispectral surveys in June, August, and October 2027.
- Initiated collaboration with USGS Northern Rocky Mountain Science Center.

Specific Objectives Met:

- Objective 1 (Field monitoring): All 24 plots sampled in all four seasons. Data quality >98%.
- Objective 2 (Remote sensing): Generated high-resolution NDVI maps for the study area covering 3 survey dates.
- Objective 3 (Soil microbiology): Processed 96 soil samples for metagenomic analysis at BSU.

2. Publications

1. Chen, S., Torres, M., & Rodriguez, E. (2027). 'Elevational gradients in alpine soil microbial diversity: A metagenomic perspective.' *Ecology Letters*, 30(4), 512-528. DOI: 10.1111/ele.14523
2. Torres, M. & Chen, S. (2027). 'Drone-based monitoring of alpine vegetation phenology.' *Remote Sensing of Environment*, 298, 113842. DOI: 10.1016/j.rse.2027.113842
3. Park, J.L., Chen, S., & Williams, K. (2027). 'Snowpack decline and its effects on alpine meadow plant community composition.' *Journal of Ecology*, 115(2), 234-249. (In review)
4. Chen, S. (2027). 'Climate change impacts on Northern Rocky Mountain ecosystems: A synthesis.' Invited chapter in 'Mountain Ecosystems Under Pressure' (Springer). (In press)
5. Rodriguez, E. & Chen, S. (2027). 'Novel soil bacteria from alpine environments: Potential for cold-adapted enzyme discovery.' *Applied and Environmental Microbiology*. (Submitted)

3. Students and Training

Graduate Students:

- Maria Gonzalez (Ph.D., Year 3) - Dissertation on alpine plant community dynamics.
- Tyler Morrison (M.S., Year 2) - Thesis on soil carbon cycling.
- Priya Sharma (Ph.D., Year 1) - Starting dissertation on microbial functional genomics.

Undergraduate Researchers:

- 6 undergraduates participated in summer field work (REU supplement pending).
- 2 undergraduates presented posters at the Idaho Academy of Sciences meeting.

Total students trained this period: 9

4. Budget Expenditures

Category	Year 2 Budget	Year 2 Spent	Cumulative Spent
Personnel	\$98,000	\$96,500	\$191,500
Equipment	\$5,000	\$4,200	\$39,200
Travel	\$14,000	\$13,800	\$25,800
Supplies	\$8,000	\$7,500	\$17,500
Indirect	\$27,000	\$13,500	\$13,500
Total	\$152,000	\$135,500	\$287,500

Remaining budget: \$197,500

Burn rate is on track for the 3-year project timeline.

5. Upcoming Milestones (Year 3)

- Complete final year of seasonal field monitoring (Sep 2027 - Aug 2028).
- Submit 2-3 additional manuscripts for publication.
- Host Idaho high school teacher workshop on alpine ecology (Summer 2028).
- Prepare and submit final project report to NSF.
- Archive all data in the Environmental Data Initiative (EDI) repository.
- Present findings at ESA 2028 Annual Meeting.